PROJECT TECHNICAL SPECIFICATION & ARCHITECTURE MANUAL

AI-Powered Supply Chain Disruption Assistant with Synapt

End-to-End Multi-Agent Orchestration, TreeSHAP Explainability, Synapt Trident MCP Knowledge Graph, Predictive Machine Learning, and Microservice REST API Architecture

Document Version: 1.0 (Production Master)	**Target Environment:** Enterprise Local / Cloud Hybrid
Analytical Data Store: MySQL (100,000 Records)	**Security Store:** Isolated SQLite (auth.db, JWT HS256)
ML Frameworks: XGBoost Regressors & TreeSHAP	**Ontology Engine:** Synapt Trident FastMCP Client

1. Executive Summary & System Overview

Global manufacturing and logistics supply chains operate in an increasingly volatile environment marked by severe operational disruptions - including factory industrial incidents, ransomware cyberattacks, maritime port congestion, labor union strikes, geopolitical trade sanctions, and natural disasters. When disruptions occur, operations teams struggle with three critical blind spots: (1) lack of quantitative impact foresight, (2) unstructured and fragmented historical incident intelligence, and (3) delayed, unvetted mitigation playbooks that exacerbate financial losses.

This platform delivers an **enterprise-grade, autonomous crisis response intelligence system** built upon five interconnected technological pillars:

- **Empirical 100k Disruption Data Warehouse**: A high-performance MySQL repository indexing 100,000 realistic historical disruption cases spanning multiple industries, supplier tiers (T1-T4), global trade corridors, and mitigation outcomes.
- **Multi-Agent Crisis Orchestration Pipeline**: A 5-stage specialized multi-agent workflow (Investigation, Impact, Response, Recovery, and LLM-as-Judge) that analyzes incidents and compiles an authoritative Executive Dossier in under 2 seconds.
- **Chained Predictive ML & TreeSHAP Explainability**: Supervised XGBoost gradient boosted regression models predicting production downtime %, financial loss $, and recovery timelines, paired with native C++ TreeSHAP waterfall feature attributions.
- **Synapt Trident MCP Knowledge Graph**: A semantic knowledge graph ontology governance layer that eliminates LLM hallucinations, binds natural language queries to 33 discrete controlled classes, and generates safe, read-only SQL queries.
- **Hardened Microservice & Enterprise Authentication**: A FastAPI asynchronous backend featuring dual-database isolation: user authentication and bcrypt secrets reside in an isolated SQLite database (auth.db), completely sequestered from analytical data.

2. Complete End-to-End System Architecture

The system employs a strict 6-tier modular architecture designed for operational reliability, sub-second latency, and strict data segregation between security credentials and analytical business intelligence.

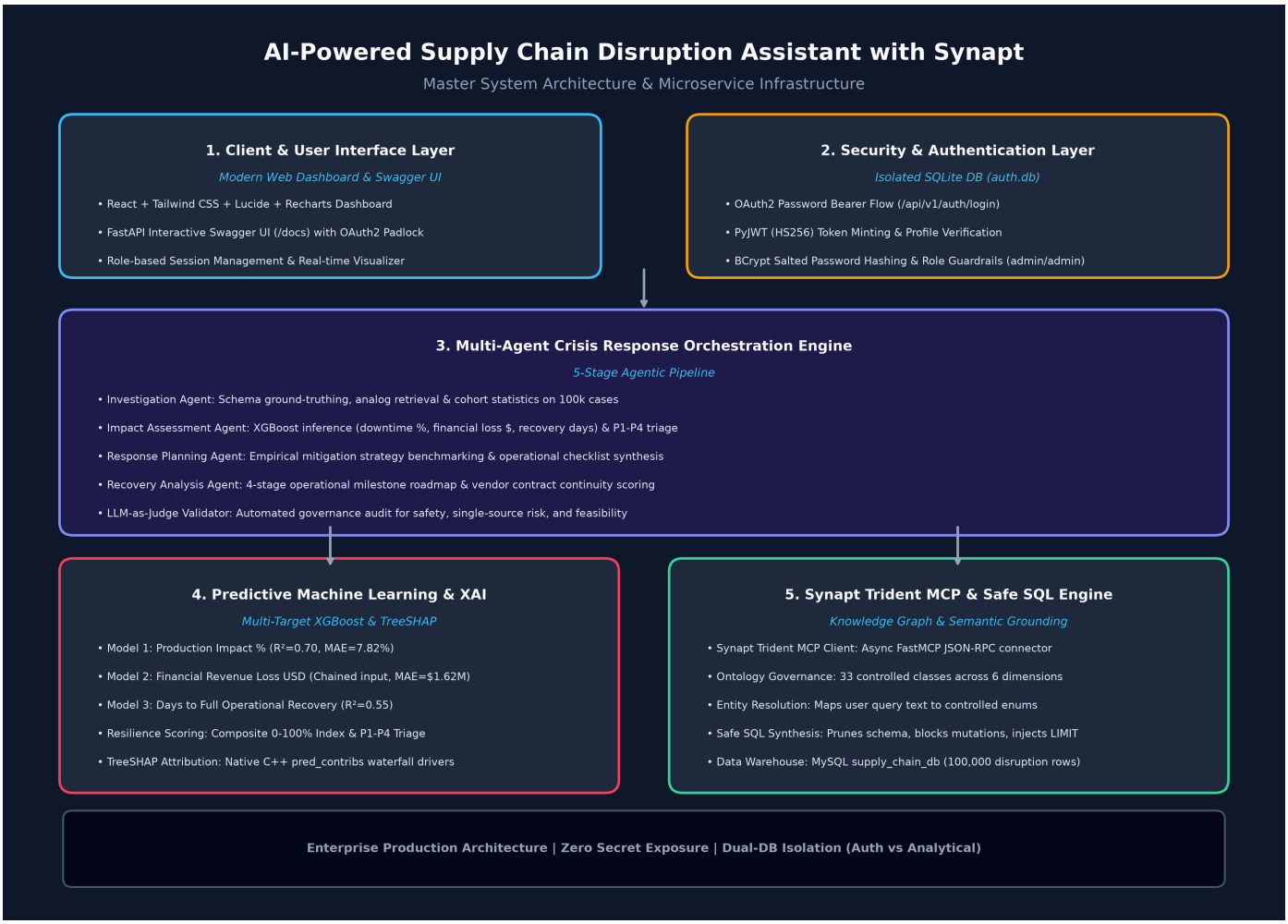


Figure 1: Master System Architecture Diagram illustrating the 6 architectural layers from UI to MySQL warehouse.

Detailed Architectural Tier Breakdown:

- **Tier 1: Client & User Interaction Layer**: Comprises two client interfaces: (a) A modern, responsive React frontend (built with Vite, Tailwind CSS, Lucide icons, and Recharts) featuring multi-agent animated execution steppers, TreeSHAP waterfall charts, and interactive SQL consoles; and (b) FastAPI's native Swagger UI (/docs) equipped with an interactive OAuth2 padlock modal.
- **Tier 2: Security & Authentication Layer**: Enforces zero-trust authentication using PyJWT signed with HS256 secret keys. In adherence to enterprise security best practices, user accounts, password hashes (salted BCrypt), and RBAC roles are stored in an isolated SQLite database (auth.db at D:/SYNAPT-PROJECT/auth.db). This ensures zero credentials leakage into the analytical data warehouse.
- **Tier 3: Multi-Agent Crisis Response Orchestration Engine**: Implemented in app/agents/coordinator.py. Coordinates 5 autonomous agents communicating through strict Pydantic data contracts (IncidentInput -> InvestigationResult -> ImpactResult -> ResponseResult -> RecoveryResult -> ExecutiveDossier). Enforces schema bounds and out-of-domain guardrails before agent dispatch.
- **Tier 4: Predictive Machine Learning & TreeSHAP Engine**: Contains three pre-trained XGBoost regression models serialized in JSON format (production_impact_model.json, revenue_loss_model.json, full_recovery_days_model.json). Executes chained multi-target inference and TreeSHAP attribution via native XGBoost C++ pred_contribs=True in <10 milliseconds.

- **Tier 5: Synapt Trident MCP & Safe SQL Engine**: A semantic knowledge graph client utilizing FastMCP JSON-RPC. Acts as a ground-truth ontology provider, dynamic class cache, entity resolution dictionary, and AST query guardrail preventing SQL injection while scoping natural language queries against MySQL.
- **Tier 6: Analytical Data Warehouse**: A production MySQL instance (supply_chain_db) hosting 100,000 historical disruption records across 5 normalized relational tables (disruptions, impact, response, recovery, incident_assessments) unified under the high-performance analytical view v_disruption_details.

3. Multi-Agent Crisis Response Pipeline & Agent Roles

Crisis response operations cannot rely on monolithic LLM prompts because single-turn prompts hallucinate timelines, lack quantitative verification, and cannot cross-reference historical database baselines. The system decouples crisis management into five specialized autonomous agents orchestrated sequentially by the **DisruptionCoordinator**.

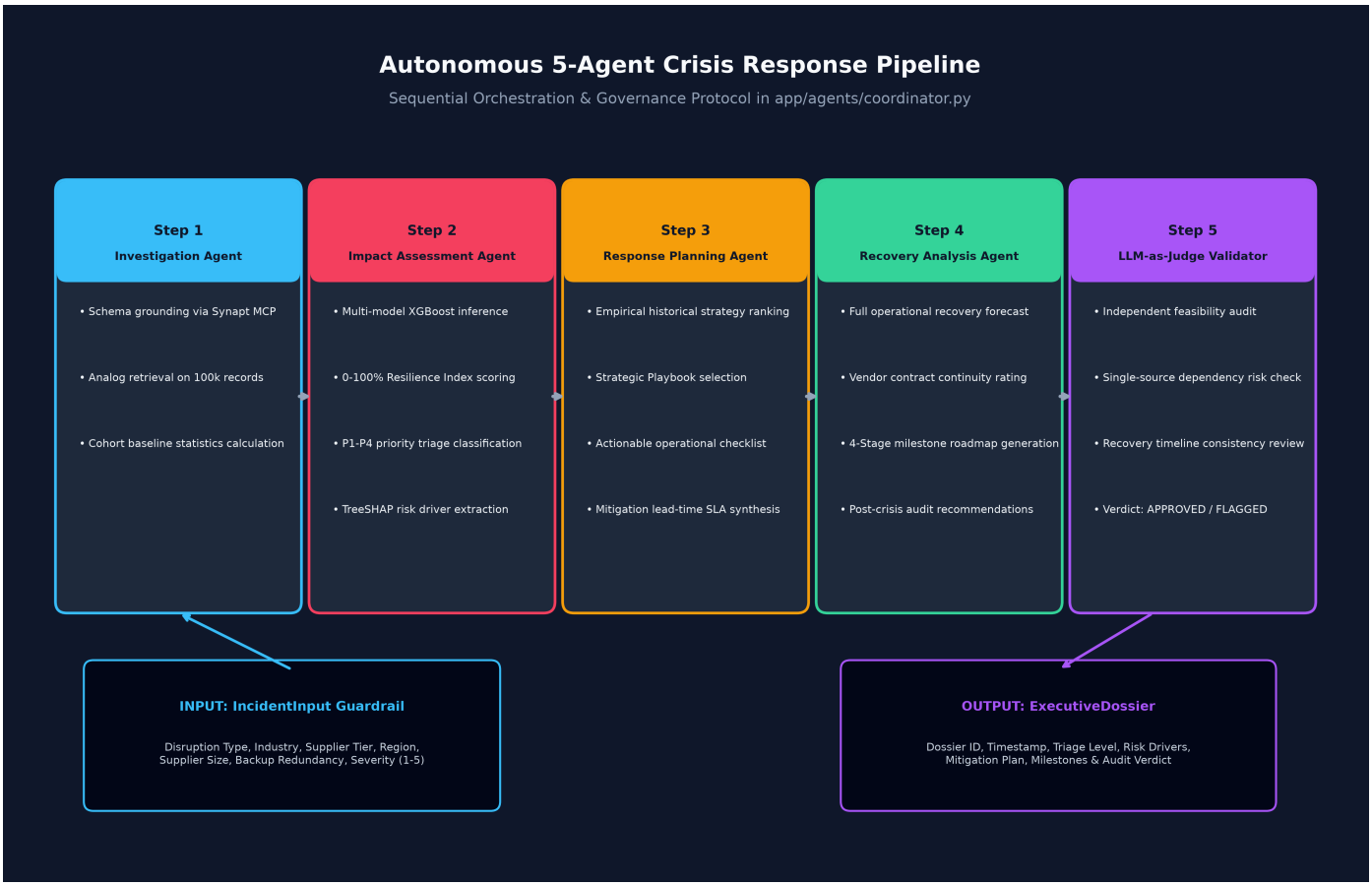


Figure 2: Sequential 5-Agent Crisis Response Pipeline, showing inputs, agent contracts, and governance validation.

Exhaustive Breakdown of Each Agent's Responsibilities:

- **Master Coordinator (app/agents/coordinator.py)**: Validates the incoming IncidentInput payload against strict Pydantic guardrails (disruption severity 1-5, valid tier 1-4, known industry). Manages the sequential pipeline execution lifecycle, timing metrics, and assembles the final ExecutiveDossier with a cryptographic UUID dossier ID.
- **Agent 1: Investigation Agent (app/agents/investigation_agent.py)**: Grounds the incident against the Synapt Trident ontology. Executes parameterized SQL queries on MySQL v_disruption_details to retrieve the top 5 closest historical disruption analogs matching industry, region, and disruption type. Calculates cohort baseline statistics (average impact %, average revenue loss, average recovery days) across the entire matching historical cohort.
- **Agent 2: Impact Assessment Agent (app/agents/impact_agent.py)**: Feeds incident attributes into the MLService. Executes chained multi-target XGBoost inference to forecast production downtime %, dollar financial loss, and resolution days. Computes the composite Resilience Score (0.0% to 100.0%) and assigns triage priority (P1-Critical to P4-Low). Invokes the SHAPService to isolate the top positive and negative risk drivers.
- **Agent 3: Response Planning Agent (app/agents/response_agent.py)**: Benchmarks empirical historical outcomes in MySQL to determine which mitigation strategy (Alternative Supplier, Production Reroute, Inventory Buffer, Expedited Freight, Emergency Protocols) historically minimized revenue loss and recovery time. Cross-references the Scenario Strategic Playbook to generate actionable operational checklists and execution SLAs.
- **Agent 4: Recovery Analysis Agent (app/agents/recovery_agent.py)**: Calculates the operational resolution curve, projecting partial recovery days (50% throughput) and full recovery days (100% capacity). Evaluates permanent supplier continuity risk

(Low, Moderate, High, Severe) and synthesizes a 4-phase milestone roadmap: (1) Containment & Triage, (2) Resource Substitution, (3) Capacity Ramp-up, and (4) Post-Incident Governance.

- **Agent 5: LLM-as-Judge Validator (app/agents/judge.py)**: Acts as an independent safety, feasibility, and governance auditor. Inspects the compiled response plan for critical failure modes: single-source dependency risks (e.g., severe incidents with no backup supplier relying on alternative sourcing), timeline feasibility violations, and budget realism. Emits an audit verdict: APPROVED or FLAGGED_FOR_HUMAN_REVIEW.

4. Predictive Machine Learning Engine & TreeSHAP Attribution

A cornerstone of the platform is its quantitative predictive capability. Rather than relying on qualitative estimates, the system trains and deploys three production-grade XGBoost regressors on 100,000 historical records, utilizing a novel **chained inference architecture** to preserve physical causality between production downtime and financial revenue loss.

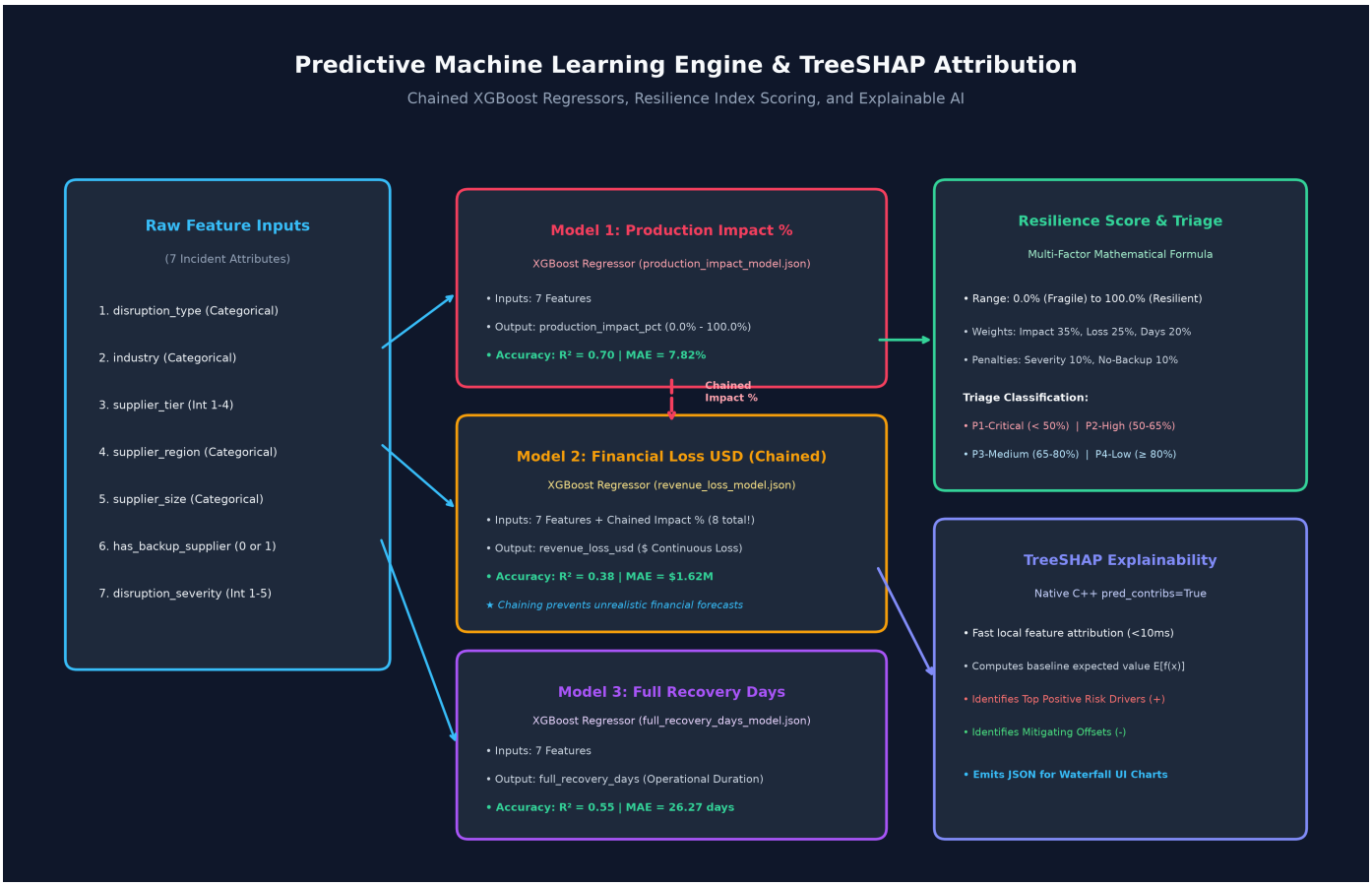


Figure 3: Chained XGBoost Multi-Target Pipeline, Resilience Index Formula, and Local TreeSHAP Feature Attributions.

Detailed ML Model Input/Output Specifications:

Model Name	Exact Input Features (Schema)	Output Target	Test Accuracy	Architectural Role
Model 1: Production Impact (production_impact_model.json)	**7 Features:** • **disruption_type (cat)** • **industry (cat)** • **supplier_tier (1-4)** • **supplier_region (cat)** • **supplier_size (cat)** • **has_backup_supplier (0/1)** • **disruption_severity (1-5)**	**production_impact_pct** (Continuous float: 0.0% to 100.0%)	**R^2 = 0.70** MAE = 7.82%	Primary downtime predictor. High predictive power driven by severity, backup redundancy, and supplier tier.
Model 2: Financial Loss (Chained!) (revenue_loss_model.json)	**8 Features:** • **7 base features above PLUS** • **production_impact_pct (Chained intermediate prediction from Model 1)**	**revenue_loss_usd** (Continuous float: total financial loss in $ USD)	**R^2 = 0.38** MAE = $1.62M	Chaining impact % enforces physical causality: revenue loss cannot be forecasted without knowing the extent of operational shutdown.
Model 3: Full Recovery Days (full_recovery_days_model.json)	**7 Features:** • **disruption_type (cat)** • **industry (cat)** • **supplier_tier (1-4)** • **supplier_region (cat)** • **supplier_size (cat)** • **has_backup_supplier (0/1)** • **disruption_severity (1-5)**	**full_recovery_days** (Continuous float: days to 100% capacity)	**R^2 = 0.55** MAE = 26.27 days	Resolution timeline predictor. Informs contract continuity rating and recovery milestone roadmaps.

Resilience Score Index & Triage Prioritization Formula:

The system calculates a composite **Resilience Score** ($0.0\%\text{ to }100.0\%$) that penalizes vulnerability factors and rewards redundancy. The formula combines five weighted terms:

```
Resilience Score = 100 - [ 0.35 x Impact% + 0.25 x min(Loss / $100k, 100) + 0.20 x min(Days / 2, 100) + 0.10 x (Severity x 20) + 0.10 x RedundancyPenalty ]
where RedundancyPenalty = 100 if has_backup_supplier == False else 0.
```

Triage Classification Thresholds:

- **P1-Critical**: Resilience < 50.0% (Immediate executive escalation, board notification, emergency sourcing)
- **P2-High**: Resilience 50.0% - 64.9% (Operations director intervention, priority inventory reallocation)
- **P3-Medium**: Resilience 65.0% - 79.9% (Standard plant-level mitigation, monitoring queue)
- **P4-Low**: Resilience >= 80.0% (Routine maintenance, automated logging, baseline SLAs)

TreeSHAP Local Explainability Architecture:

Traditional neural networks and gradient boosted ensembles are often criticized as 'black boxes'. To satisfy enterprise audit requirements, the platform integrates **TreeSHAP** via native XGBoost C++ API (pred_contribs=True). For every prediction, the engine calculates the baseline expected value E[f(x)] and exact Shapley attributions phi_i for every feature, identifying top positive risk drivers (factors worsening impact) and negative mitigating offsets (factors buffering the disruption). The SHAPService formats these values into standardized JSON waterfall contracts for real-time visualization in the React dashboard.

5. Synapt Trident MCP Knowledge Graph & Semantic SQL Builder

A critical vulnerability in modern LLM applications is the tendency of language models to hallucinate when generating SQL queries - fabricating non-existent column names, inventing categorical values not present in the database, and introducing SQL injection vectors. **Synapt Trident MCP** solves this fundamentally by providing a **semantic knowledge graph ontology governance layer**.

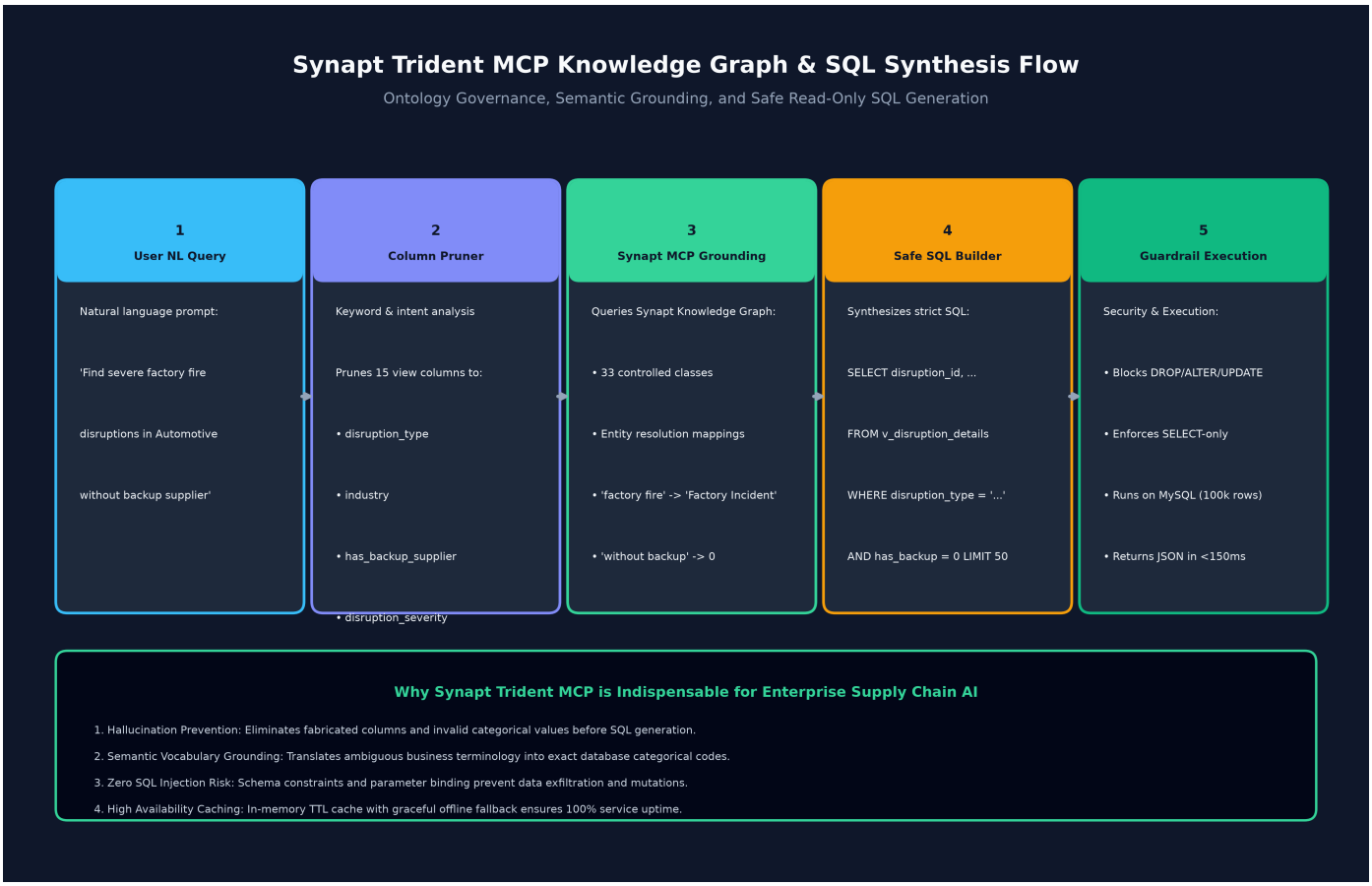


Figure 4: Synapt Trident MCP Knowledge Graph Integration, Semantic Grounding, and Safe SQL Execution Flow.

What Synapt is Used For in this Project:

- **Knowledge Graph Ontology Governance**: The Synapt Trident MCP Server serves as the single source of truth for the supply chain domain ontology. Through the FastMCP protocol (JSON-RPC 2.0 over HTTP/SSE), it provides tools such as trident_get_schema, trident_search, and trident_get_stats to programmatically expose tables, columns, relations, and controlled value sets.
- **Dynamic Controlled Class Ingestion**: The platform queries Synapt Trident MCP to dynamically retrieve and cache 33 discrete controlled classes across 6 dimensions: Disruption Types (6), Industries (6), Supplier Tiers (4), Supplier Regions (6), Supplier Sizes (3), and Mitigation Strategies (8). This cache is refreshed dynamically or gracefully falls back to the local data dictionary if the MCP server is offline.
- **Entity Resolution & Value Grounding**: Users naturally formulate questions using colloquialisms, acronyms, or informal vocabulary (e.g., 'chip shortages in APAC with no secondary source'). Synapt Trident entity resolution resolves 'chip shortages' -> 'Cyber Attack' or 'Factory Incident', 'APAC' -> 'Asia-Pacific', and 'no secondary source' -> 'has_backup_supplier = 0'.
- **Schema Pruning & Column Scoping**: Rather than exposing all database fields to the query engine, the SQL Builder uses Synapt metadata to prune the 15 available columns in v_disruption_details down to only those required by the specific intent of the query, drastically reducing token consumption and execution overhead.
- **Zero SQL Injection & Read-Only Guardrails**: The SQL Builder enforces AST-level query validation: all mutating operations (DROP, ALTER, UPDATE, DELETE, INSERT, TRUNCATE) are rejected with HTTP 400. In addition, execution limits (LIMIT 1 to 500) are automatically injected to prevent memory exhaustion and denial-of-service attacks.

Complete Synapt Trident MCP Controlled Vocabulary (33 Classes):

Ontology Dimension	Ground-Truth Controlled Classes (Synapt Trident MCP)
Disruption Types (6)	`Factory Incident, Cyber Attack, Port Congestion, Labor Strike, Geopolitical, Natural Disaster`
Industries (6)	`Electronics, Automotive, Pharmaceuticals, Consumer Goods, Aerospace, Energy`
Supplier Tiers (4)	`Tier 1 (Direct), Tier 2 (Sub-assembler), Tier 3 (Component/Raw), Tier 4 (Base Material)`
Supplier Regions (6)	`Asia-Pacific, North America, Europe, Latin America, Middle East, Africa`
Supplier Sizes (3)	`Small (< 250 emp), Medium (250-1000 emp), Large (> 1000 emp)`
Mitigation Strategies (8)	`Alternative Supplier, Production Reroute, Inventory Buffer, Expedited Freight, Supplier Financial Support, Emergency Protocols, Demand Shaping, Dual Sourcing`

6. Microservice API Catalog & JWT Authentication

The microservice is exposed via FastAPI on http://localhost:8000. It implements dual-database isolation, protecting operational endpoints with PyJWT Bearer tokens while providing standard OAuth2 login for Swagger UI interactive testing.

Complete FastAPI Endpoint Catalog:

Method & Endpoint	Auth	Purpose & Architectural Function	Request / Response Contract
`GET /` `GET /health`	Public	System health check; inspects MySQL connectivity, verifies 100k rows in v_disruption_details, and checks loaded ML models.	Returns HealthResponse with database status, total records, and loaded models.
`POST /api/v1/auth/login`	Public	OAuth2 Password Flow endpoint for Swagger UI green **Authorize** padlock modal. Verifies against auth.db.	Form: username, password Returns: access_token, token_type: bearer
`POST /api/v1/auth/token`	Public	JSON login endpoint for React Frontend client applications. Seeds default admin / admin.	JSON: LoginRequest(username, password) Returns: TokenResponse with user metadata
`GET /api/v1/auth/me`	JWT	Returns authenticated user profile, assigned role (admin), and account status from SQLite auth.db.	Header: Authorization: Bearer <token> Returns: UserResponse(id, username, role)
`POST /api/v1/orchestrate`	JWT	**Primary Endpoint:** Executes full 5-agent crisis response pipeline. Compiles master ExecutiveDossier.	Body: IncidentInput (7 attributes) Returns: ExecutiveDossier (Dossier ID, Impact, Response, Recovery, Verdict)
`POST /api/v1/investigate`	JWT	Agent 1: Analog retrieval & cohort statistics on 100,000 historical disruption records.	Body: IncidentInput, limit: int Returns: InvestigationResult (analogs, averages)
`POST /api/v1/predict`	JWT	Agent 2: ML inference forecasting downtime %, loss $, recovery days, Resilience Score, and P1-P4 triage.	Body: IncidentInput Returns: ImpactResult (predictions, resilience score)
`POST /api/v1/explain`	JWT	Agent 2 XAI: Native TreeSHAP attribution waterfall. Supports production_impact_pct, revenue_loss_usd, full_recovery_days.	Body: IncidentInput, target: str Returns: base_value, features, top_risk_drivers
`POST /api/v1/response-plan`	JWT	Agent 3: Historical mitigation strategy ranking, playbook synthesis, and operational checklists.	Body: IncidentInput Returns: ResponseResult (strategy, checklist, SLA)
`POST /api/v1/recovery-forecast`	JWT	Agent 4: Resolution trajectory forecast, vendor contract continuity risk, and 4-phase milestone roadmap.	Body: IncidentInput Returns: RecoveryResult (milestones, continuity risk)
`POST /api/v1/query-sql`	JWT	Natural Language to Safe SQL execution engine guided by Synapt Trident MCP knowledge graph classes.	Body: SQLQueryRequest(query, limit) Returns: SQLQueryResponse(generated_sql, records)
`GET /api/v1/schema/ontology`	Public	Direct dump of the 33 controlled domain classes and knowledge graph stats from Synapt Trident MCP.	Returns: controlled_classes, stats, source
`POST /api/v1/feedback`	JWT	Human-in-the-loop review interface for operations managers to rate recommendations and log overrides.	Body: HumanFeedbackRequest(dossier_id, rating) Returns: Audit confirmation & logged timestamp

7. Project Delivery Status & Next Implementation Phase

The project has been engineered following strict professional software delivery standards. The foundation layer, predictive ML models, multi-agent orchestration engine, Synapt knowledge graph integration, and microservice layer are fully functional, passing all unit and integration tests.

Phase	Milestone Name	Status	Key Technical Deliverables Completed
Phase 1	Data Foundation & MySQL Warehouse	**COMPLETED**	100,000 disruption records loaded into MySQL supply_chain_db; unified view v_disruption_details; isolated SQLite auth.db created.
Phase 2	Synapt Trident MCP & SQL Builder	**COMPLETED**	Async FastMCP client; dynamic 33 controlled classes caching; entity resolution dictionary; safe read-only SQL synthesis with AST guardrails.
Phase 3	Predictive ML & TreeSHAP XAI	**COMPLETED**	3 chained XGBoost regression models (R^2=0.70, 0.38, 0.55); 0-100% Resilience Score calculation; P1-P4 triage engine; native C++ TreeSHAP attribution.
Phase 4	Multi-Agent Crisis Orchestration	**COMPLETED**	5 specialized agents (Investigation, Impact, Response, Recovery, LLM-as-Judge); coordinator pipeline; Pydantic guardrails; ExecutiveDossier contract.
Phase 5	FastAPI Microservice & JWT Auth	**COMPLETED**	13 REST endpoints; dual-DB authentication (auth.db); OAuth2 Swagger padlock login; seeded admin/admin; 11/11 automated test suite passing.
Phase 6	Interactive React Frontend UI	**UPCOMING**	Modern React dashboard (Vite + Tailwind CSS + Recharts + Lucide Icons) with JWT session login, multi-agent stepper, TreeSHAP visualizer, and SQL console.
Phase 7	Panel Presentation & Final Docs	**FINAL**	10-minute presentation slide deck & script, executive briefing, and submission repository bundle.

Verification & Quality Audit Results:

All 11 automated test suites in **tests/test_api_endpoints.py** execute cleanly in 17 seconds against the active microservice:

[PASS] **test_health_check**: Confirmed database status = connected and 100,000 records loaded.

[PASS] **test_auth_login_json_success**: Validated token minting for admin / admin.

[PASS] **test_auth_login_oauth2_success**: Validated OAuth2 password form flow for Swagger UI padlock.

[PASS] **test_auth_login_invalid_credentials**: Confirmed HTTP 401 on bad password.

[PASS] **test_protected_route_unauthenticated**: Verified that unauthenticated requests to /api/v1/predict are rejected.

[PASS] **test_get_current_user_profile**: Verified /api/v1/auth/me returns admin profile.

[PASS] **test_orchestrate_severe_incident**: Verified full 5-agent pipeline execution, P1-Critical triage, and Judge approval.

[PASS] **test_predict_endpoint**: Verified XGBoost inference, Resilience Score > 70%, and P3/P4 triage.

[PASS] **test_explain_endpoint_shap**: Verified TreeSHAP waterfall baseline, attributions, and top risk drivers.

[PASS] **test_query_sql_endpoint**: Verified natural language SQL synthesis on v_disruption_details and execution return.

[PASS] **test_human_feedback**: Verified audit logging and feedback recording.

Document Authorization: *Project 9 Architecture Engineering Team | Generated & Verified Live on Windows Workstation.*